

## SECTION 4 - RAMP FILL



**MAKE SURE THAT THE FOLLOWING ACTIONS ARE TAKEN BEFORE STARTING THIS PROCEDURE TO PREVENT POTENTIAL FATAL INJURY !!!**

- **REVIEW AND FULLY UNDERSTAND ALL SUPERCONDUCTING MAGNET PORTIONS OF SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.**
- **FULLY COMPLY WITH ALL REQUIRED ITEMS FOR THIS PROCEDURE IN SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-4, MAGNET & CRYOGEN SERVICE SAFETY REQUIREMENTS.**
- **HAVE ALL “WORK ASSISTANTS” OR “WORK OBSERVERS” COMPLY WITH SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS, SECTION 2-5, BUDDY SYSTEM REQUIREMENTS & CERTIFICATION.**

### **IMPORTANT !!!**

Before starting Ramp Fill open Valve V3 to check for gas flow and exhaust plumbing cooling. No gas flow / cooling is an indication of ice in the Recondenser or exhaust plumbing which must be cleared before starting Ramp Fill. See REPLACEMENT / MAINTENANCE, Section 2, Ice Removal.

#### **4-1 INTRODUCTION**

The Main and Bucking Coils of the GE 0.7T OpenSpeed Magnet are in a horizontal plane to generate a vertical magnetic field. Because of this configuration, it is essential that the magnet helium level remain between 95% and 98% during the ramp to be in contact with the Top Bucking Coil. Failure to maintain this condition will result in a cooling loss of the Top Bucking Coil and could result in a quench.

The ramp / fill procedure maintains the required helium level during the ramp through the controlled replenishment of liquid helium to compensate for loss through the Main Leads and Vent Valve V3.

Because of the complexity and simultaneous tasks of the ramp / fill procedure, two trained and certified persons are required. Person 1 performs the ramp while person 2 simultaneously performs the fill.

#### 4-1 INTRODUCTION ( continued )

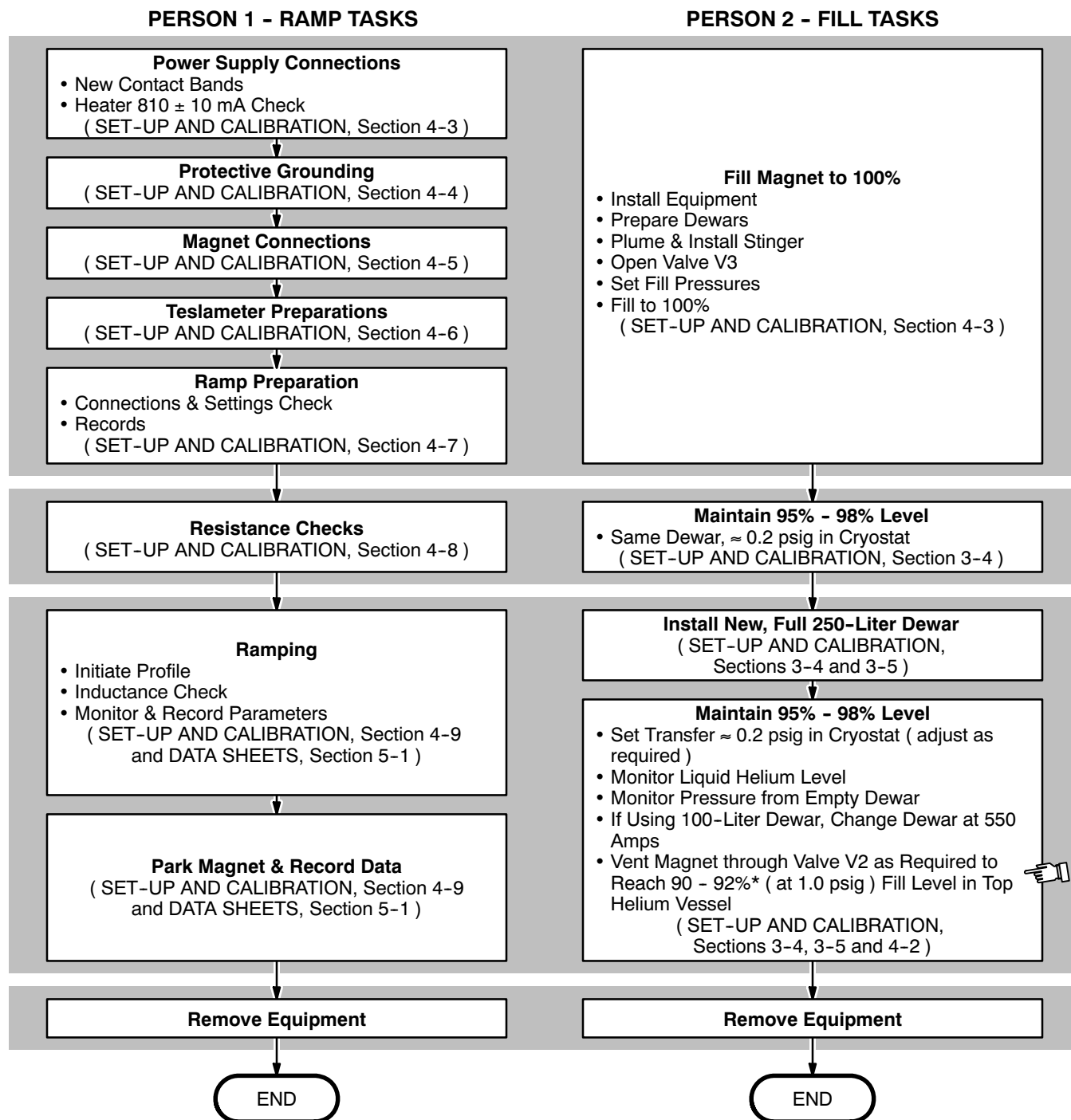
Before starting these procedures, review and fully understand all superconducting magnet portions of 2301164PRE, *MR Magnet - Safety Requirements*, found in the SAFETY REQUIREMENTS tab of this manual. Fully comply with that document's Section 2-4, Magnet & Cryogen Service Safety Requirements. Have all "Work Assistants" or "Work Observers" comply with that document's Section 2-5, Buddy System Requirements & Certification.

Table 4-1 outlines the ramp / fill process steps. Illustration 4-1 highlights the tasks and sequence for each person. Complete DATA SHEETS, Table 5-2, Magnet Ramp / Fill Profile, for both magnet ramp up and ramp down.

TABLE 4-1  
RAMP / FILL PROCESS STEPS

| Prerequisites                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• Magnet Heaters must be stabilized.</li> <li>• Perform Power Supply checks to verify functional ramp supply.</li> <li>• Magnet Monitor must be functional and put into "FILL" mode.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| The process                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <ul style="list-style-type: none"> <li>• Vent magnet down to .5 psig.</li> <li>• If the Upper Vessel helium level is &lt; 80%, top off the magnet to at least 95%. Valve V3 should be open when helium level is greater than 70%.</li> <li>• If the Upper Vessel helium level is &gt; 80%, proceed to next step.</li> <li>• Install Protective Grounding Circuit.</li> <li>• Insert the Main Lead Extensions.</li> <li>• Install the Hold-down Tool, but do not fully tighten.</li> <li>• Insert the fill line Stinger and begin filling the magnet.</li> <li>• Tighten the Hold-Down Tool.</li> <li>• When the Upper Vessel helium level is &gt; 90%, do resistance checks.</li> <li>• When the resistance checks pass, change to a new 250-liter Dewar.</li> <li>• Park magnet at 29.793 MHz. This will account for the frequency drop when the supply is ramped down and the frequency increase when the magnet is shimmed. Specification for 0.7T OpenSpeed Magnet is <math>29.803 \pm 0.01</math> MHz.</li> <li>• Replace Power Supply Jumper Cable before removing the Main Lead Extensions.</li> <li>• After the supply is turned off, continue to fill while removing the Hold-Down Tool and Main Lead Extensions. This will give a higher helium level for magnet turnover to the customer. Hold-Down Tool and Main Lead Extensions should be removed quickly.</li> <li>• Put the Magnet Monitor back into normal operating mode.</li> </ul> |

## 4-1 INTRODUCTION ( continued )



\* 80 - 85% at sites scanning > 10 patients / day.

**RAMP / FILL PROCESS**  
ILLUSTRATION 4-1

#### 4-1 INTRODUCTION ( continued )

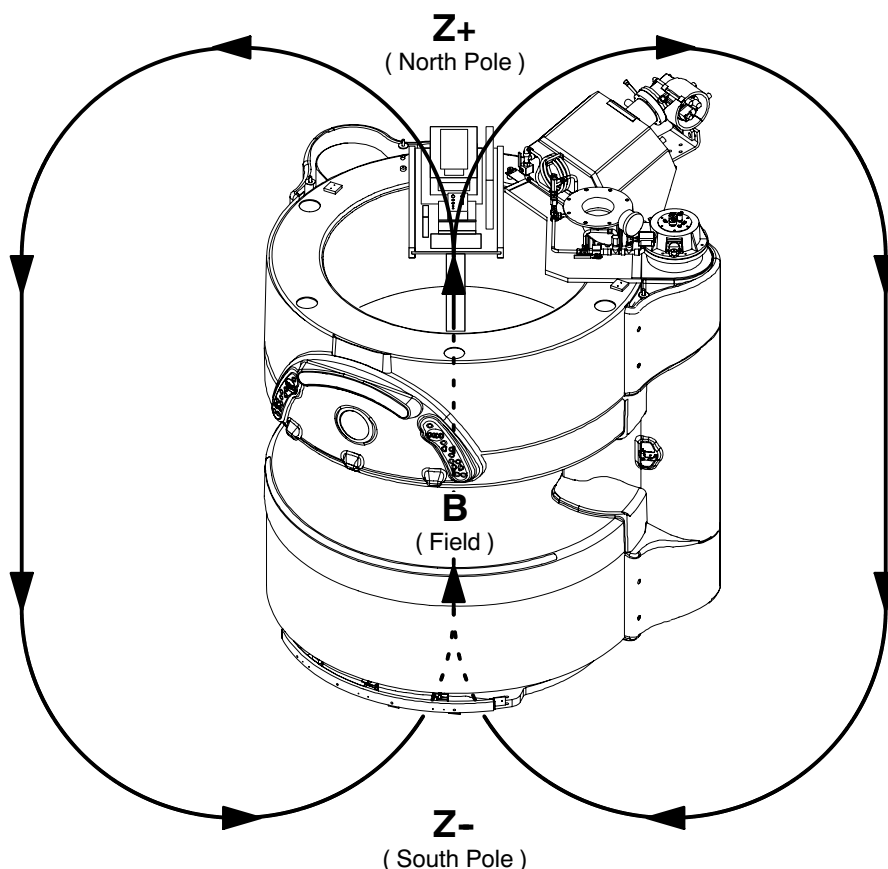
### IMPORTANT !!!

It is essential that the following conditions are met to prevent a 250-liter helium Dewar from running dry during the ramp. Approximately 150 liters of helium will be used under these conditions.

- Follow the ramp profile. Do not allow the ramp voltage to decay during each step of the ramp.
- Make sure contact resistance ( < 150 mV @ 750 amps ) is well within limits to minimize internal heat generation.
- Connect a new 250-liter helium dewar after resistance checks ( at start of ramp ).
- Maintain a helium level of 95% - 98% in the Top Helium Vessel during the ramp. Vessel pressure should be between 0.35 psig and 0.6 psig to maintain this helium level.
- Verify that magnet Temperature Control Units ( TCU's ) have stabilized for at least 6 hours at a temperature of 28°C.

#### Note

The field polarity of a normally ramped magnet is shown in Illustration 4-2.



FIELD POLARITY OF A NORMALLY RAMPED MAGNET  
ILLUSTRATION 4-2

## 4-1 INTRODUCTION ( continued )

**WARNING!**

MAKE SURE THE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON BEFORE STARTING THIS PROCEDURE. THE ODORLESS, INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE WILL DISPLACE OXYGEN AND CAUSE ASPHYXIA UNLESS SAFELY VENTED AWAY FROM THE MAGNET ROOM.

SECURE THE MAGNET ROOM DOORS IN THE OPEN POSITION BEFORE CHANGING MAGNETIC FIELD.

REVIEW AND FOLLOW SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.

## 4-2 HELIUM FILL DURING RAMP - PERSON 2

**CAUTION**

Continuously observe the magnet vessel pressure and upper vessel helium level during the ramp / fill process. A portable helium meter should be used.

**Note**

Portable Helium Meter 2270571 is calibrated for the GE 0.7T OpenSpeed magnet. However, any helium meter calibrated for a LCC or Florence-manufactured magnet can be used if the meter level is not allowed to drop below 97% during the ramp / fill process.

1. Open Valve V2 ( identified in Illustration 4-3 ) and vent the magnet down to < 0.5 psig. Close V2.

**CAUTION**

Shield the pump-out port on the transfill line from cold helium gas that can harden the pump-out port o-ring and cause the line to go "soft."

2. Assemble the equipment, a new Dewar and cylinder of helium gas. Plume and insert the Stinger after the Main Lead Extensions and Hold-Down Tool are installed.

#### 4-2 HELIUM FILL DURING RAMP - PERSON 2 ( continued )

3. Open Shield Vent Valve V3, identified in Illustration 4-3. Fill the magnet to 100% in conformance with SET-UP AND CALIBRATION, Section 3-4, Helium Precool.

#### Note

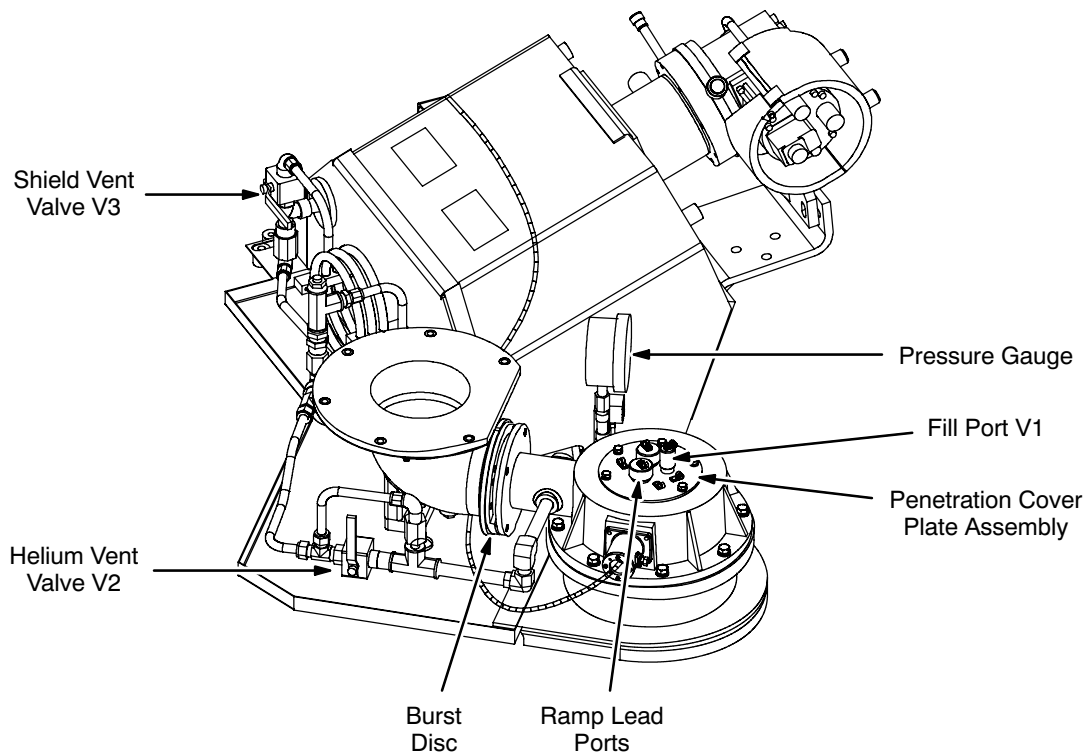
Leave Valve V3 open sufficiently to maintain 0.2 psig helium transfer pressure during the entire ramping procedure to prevent pressure build-up.

4. Maintain a 95 - 98% helium level during resistance checks. Nominal Cryostat pressure is 0.5 psig for maintaining this helium level.
5. Change to a new, full 250-liter Dewar at the start of ramp in conformance with SET-UP AND CALIBRATION, Section 3-6, Changing Helium Dewars.

#### IMPORTANT !!!

It is essential that the magnet is at 98% liquid helium level and a full 250-liter dewar is installed at the start of the ramp to prevent the Dewar from running dry during the ramp.

6. Adjust pressure from a helium gas cylinder connected to the Dewar and then to the Cryostat to produce a ~ 0.5 psig Cryostat pressure. Make minor adjustments to Dewar pressure as required to maintain between 95 - 98% liquid helium level during ramp.



**V2 AND V3 VALVE LOCATIONS**  
ILLUSTRATION 4-3

**4-2 HELIUM FILL DURING RAMP - PERSON 2 ( continued )****CAUTION**

Always maintain a positive pressure differential between the Dewar and the Cryostat to prevent any backflow of helium into the Dewar.

**WARNING!**

ALLOWING A HELIUM DEWAR TO EMPTY DURING RAMPING WILL CAUSE WARM GAS TO ENTER THE MAGNET AND CAUSE A QUENCH. CONTINUOUSLY MONITOR THE DEWAR AND MAGNET FOR A “WHISTLING” SOUND AND MAGNET PRESSURE FOR AN INCREASE; THESE INDICATE THE DEWAR IS EMPTY. WHEN AN EMPTY DEWAR IS SUSPECTED, IMMEDIATELY SHUT OFF THE DEWAR STINGER AND DISCONTINUE THE RAMP UP, MAINTAINING A CONSTANT RAMP CURRENT.

7. Continuously monitor the magnet for a noticeable pressure increase and the Dewar Stinger for a whistling sound indicating that the Dewar is empty. If the Dewar is empty, have Person 1 park the magnet, turn the Power Supply off and change Dewars in conformance with SET-UP AND CALIBRATION, Section 3-6, Changing Helium Dewars.
8. Once the magnet is ramped and parked, close V3. Then open valve V2 and vent the magnet as required to reach a 90% fill level in the Top Helium Vessel.

**CAUTION**

Fill the Top Helium Vessel only to the 90 - 92% level ( 80 - 85% at sites scanning > 10 patients / day ). Fill levels beyond 92% can result in magnet over-pressurization.

9. Immediately discontinue helium transfer. Remove all helium fill equipment in conformance with SET-UP AND CALIBRATION, Section 3-5, Liquid Helium Fill.

### 4-3 RAMP POWER SUPPLY CIRCUIT CONNECTIONS - PERSON 1

#### Introduction

At least once a year calibrate the Ramp Power Supply in conformance with FUNCTIONAL CHECKS, Section 8, ESS7.5-1000-2-D-1236 Main Power Supply Calibration / Check.

Make sure the Ramp Power Supply is installed and checked out in conformance with the vendor manual supplied with the unit before making any Power Supply connections to the magnet. See INTRODUCTION, Section 2, Vendor Manual Matrix for vendor manual numbers.

SCHEMATICS / INTERCONNECTS, Section 1-1, Magnet System Interconnects, shows the cable numbers and interconnect patterns covered in this Section.

The Main Coil Power Cables, Switch Heater Cables and Voltage Sense Leads are provided as part of the Ramp Cart / Cable Kit ( 1000 Amp ) 2180589.

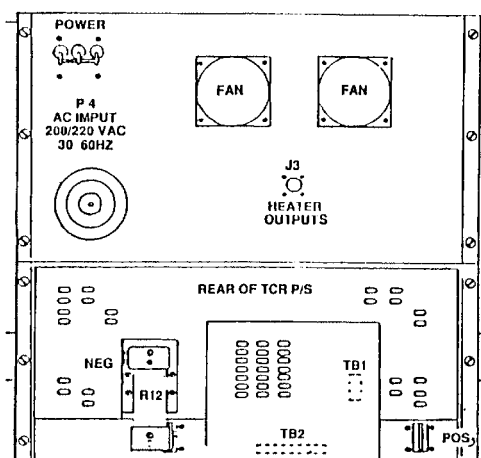
Use Main Lead Extensions 46-294204G1 from Magnet Ramping Equipment Kit 46-260703G6.

#### Procedure

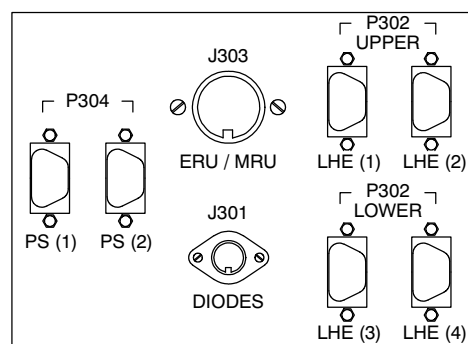
1. Verify that input power to the Main Coil Power Supply is disconnected.
2. Connect "P3" on Switch Heater Cable 46-281667G1 to Heater Outputs "J3" on the rear of the Main Coil Power Supply Cabinet ( MS6-A1 ). See Illustration 4-4.

#### Note

The magnet incorporates a "B0 Coil" to decrease moving-metal sensitivity of the magnet field. The B0 Coil Heater is energized through the Ramp Power Supply's Axial Shim Heater Circuit. The Ramp Power Supply's Axial Shim Heater must remain on throughout the Rampdown procedure in order to quench the B0 Coil field.



Power Supply ( Rear )



Connector Panel

MAGNET POWER SUPPLY OUTPUT CONNECTIONS AND MAGNET CONNECTOR PANEL

ILLUSTRATION 4-4



**4-3 RAMP POWER SUPPLY CIRCUIT CONNECTIONS - PERSON 1 ( continued )**

3. Connect "J304" on the Switch Heater Cable to "P304" on the Magnet Connector Assembly ( MS1-A3, A1 ). See Illustration 4-4.

**Note**

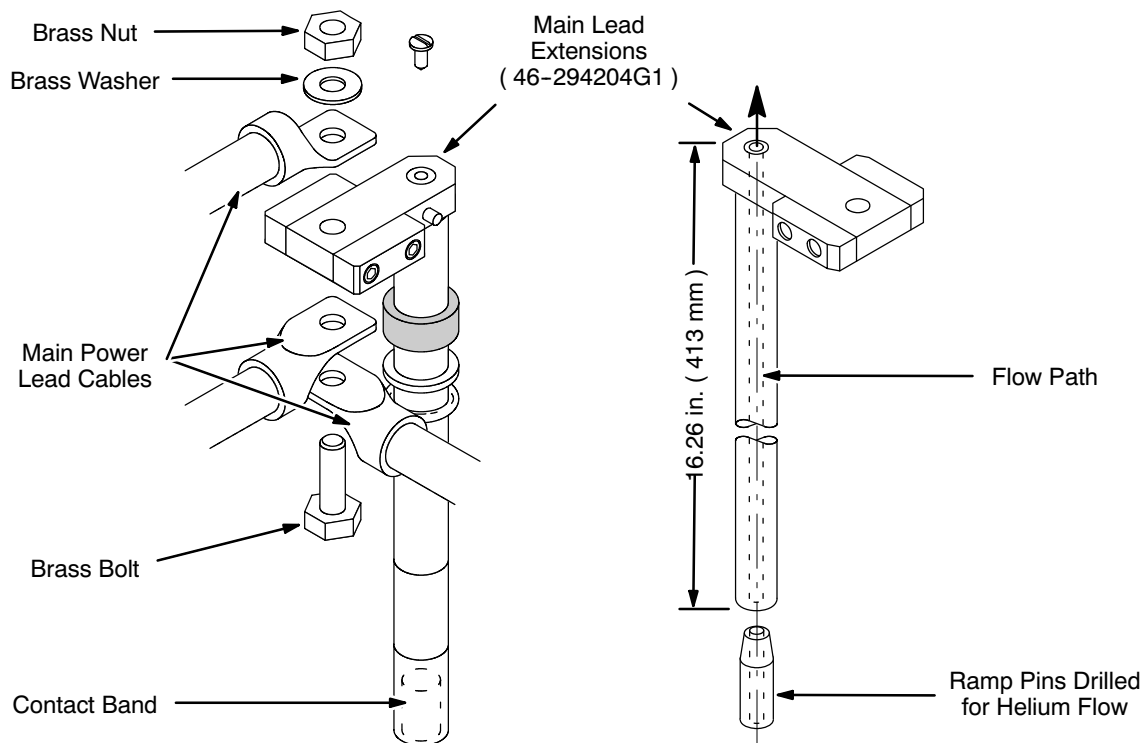
Make sure the nuts installed in Steps 4 and 6 below are tightened sufficiently to prevent a high resistance contact. Connect the red cables to the Positive ( + ) Buss Bar and the black cables to the Negative ( - ) Buss Bar on the Main Power Supply. For magnets ramped with normal polarity make sure the red ( + ) cables are connected to the "+" Magnet Ramp Port and the black ( - ) cables are connected to the "-" Magnet Ramp Port.

4. Connect the Main Coil Power Cables ( "+" Red: 46-260723G1; "-" Black: 46-260723G2 ) to the Main Power Supply Buss Bars with the .375 inch brass nuts, bolts and washers provided in the Magnet Ramping Equipment Kit ( 46-260703G4 ).

**Note**

Three pairs of Main Coil Power Cables must be used when ramping the GE 0.7T Open-Speed magnet.

5. Replace the Contact Bands on the Main Lead Extension ( 46-294204G1 ) according to REPLACE-  
MENT / MAINTENANCE, Section 8, Main Lead Extension Contact Band Replacement. Make sure the Gas Flow Holes are not blocked.
6. Connect the other end of the Main Power Cables to the Main Lead Extensions with the brass nuts, bolts and washers provided. See Illustrations 4-5 and 4-10.

**4-3 RAMP POWER SUPPLY CIRCUIT CONNECTIONS - PERSON 1 ( continued )**

**HARDWARE MOUNTING ON MAIN LEAD EXTENSIONS**  
ILLUSTRATION 4-5

#### 4-4 PRE-RAMP PROTECTIVE GROUNDING CIRCUIT CONNECTIONS - PERSON 1

**WARNING!**

**MAKE SURE THE POWER SUPPLY IS ALWAYS COMPLETELY GROUNDED IN CONFORMANCE WITH THE FOLLOWING PROCEDURE BEFORE INSERTING OR EXTRACTING ANY MAIN LEAD EXTENSION TO THE MAGNET ( DURING RAMP UP, RAMP DOWN OR REPARKING MAGNET ). GROUNDING WILL PREVENT THE REMOTE POSSIBILITY OF A HIGH VOLTAGE BETWEEN THE MAIN LEAD EXTENSION AND THE CRYOSTAT GROUND IF AN UNLIKELY QUENCH AND ARC / SHORT TO THE CRYOSTAT OCCUR WHEN THE MAIN LEAD EXTENSION IS INSERTED / EXTRACTED.**

### **IMPORTANT !!!**

Requirements during ramping:

- Always install the Pre-Ramp Protective Grounding Circuit and verify Main Lead Extension grounding to Cryostat ground before inserting or extracting any Main Lead Extension.
- Once the Main Lead Extensions are inserted, disconnect and remove the Power Supply Jumper Cable from the power supply while wearing non-absorbant leather gloves with NO holes / tears. This removes the short across the power supply and enables magnet ramping up / down.
- Once the magnet is parked and the power supply turned off — but before removing the Main Lead Extensions — reconnect the Power Supply Jumper Cable's twist connectors to the power supply while wearing non-absorbant leather gloves with NO holes / tears. This re-establishes full power supply grounding.
- After ramping up / down, when the power supply has been turned off and the Main Lead Extensions have been extracted, remove the Power Supply Jumper and Grounding Cables.

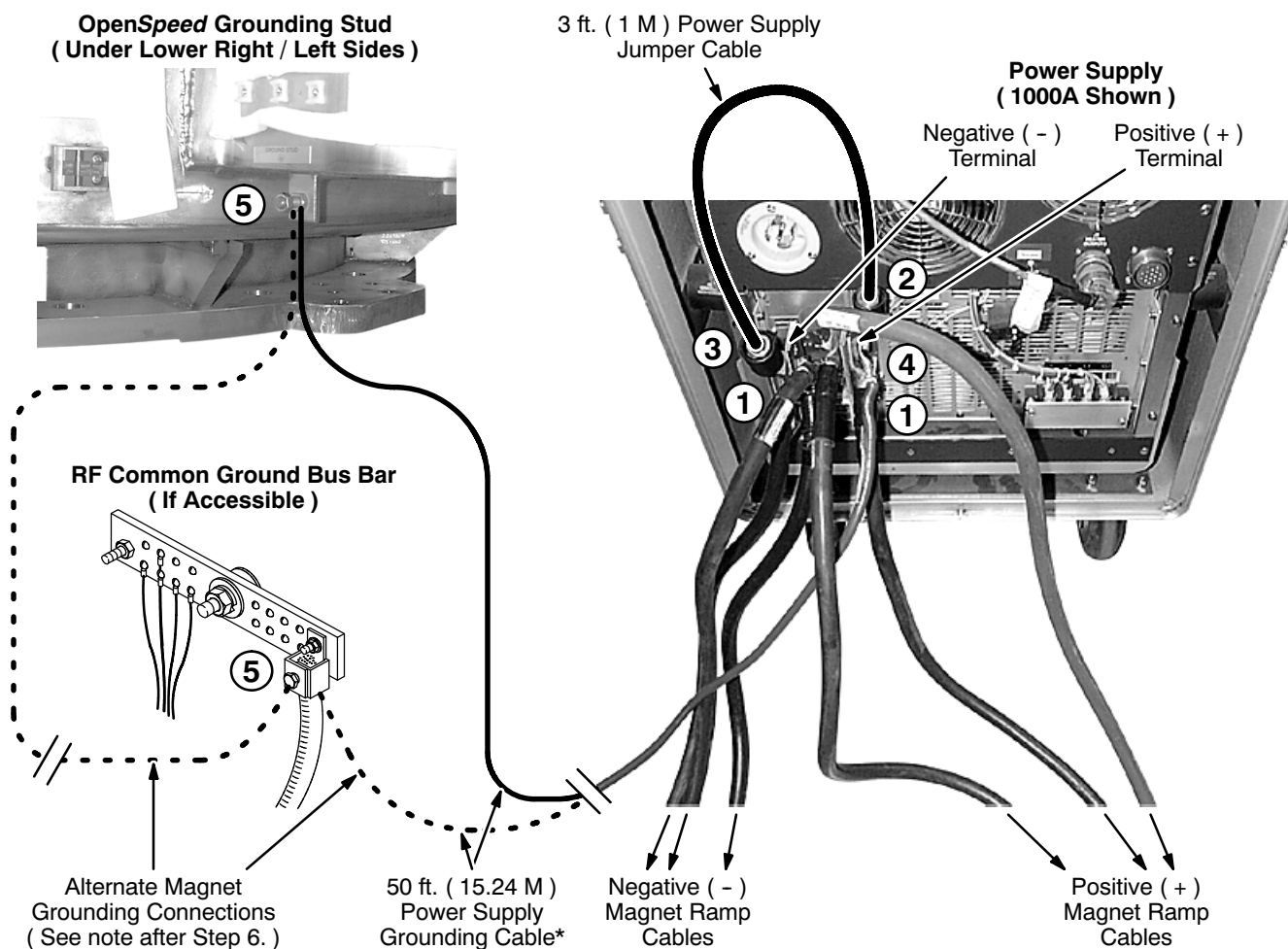
### **Equipment**

- .375-16 x 1.75 inch Brass Screws, Washers and Nuts ( 2 )
- 3 ft. ( 1 M ) Power Supply Jumper Cable
- 50 ft. ( 15.24 M ) Power Supply Grounding Cable

#### 4-4 PRE-RAMP PROTECTIVE GROUNDING CIRCUIT CONNECTIONS - PERSON 1 ( continued )

##### Procedure

1. Connect the Magnet Ramp Leads to the Ramp Power Supply in the standard configuration and using the .375-16 x 1.75 inch brass screws, washers and nuts. See Illustration 4-6, label "1".

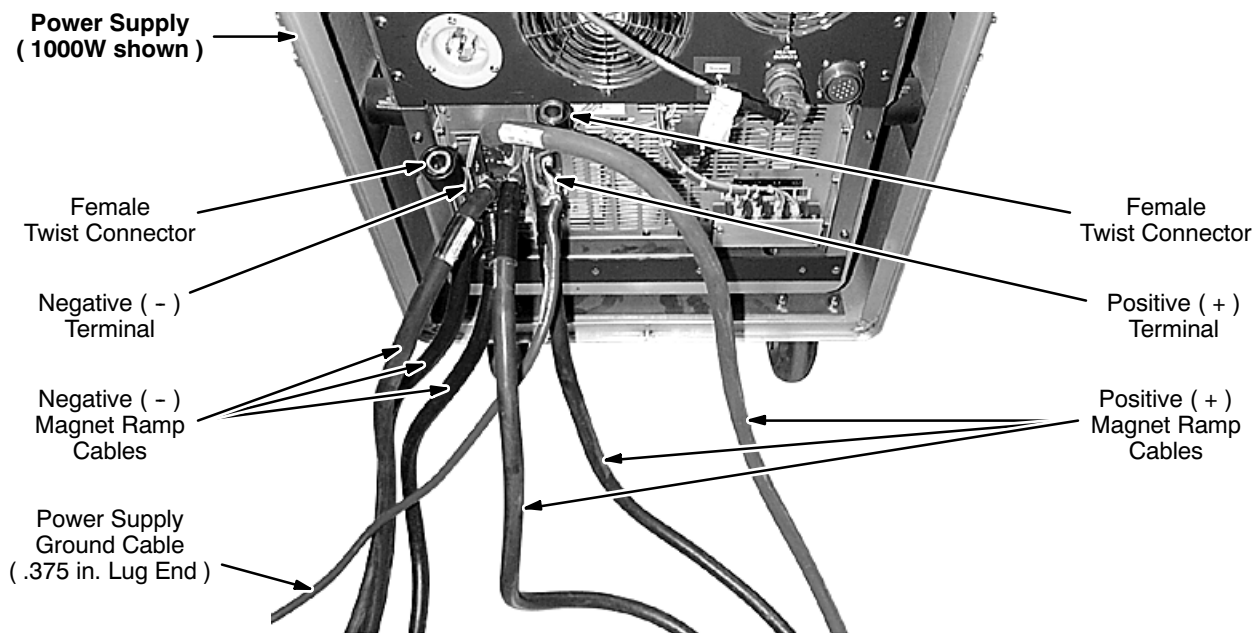


\* 3/8 inch lug at Power Supply; .5 inch lug at Magnet.

**PRE-RAMP PROTECTIVE GROUNDING CIRCUIT CONNECTIONS**  
ILLUSTRATION 4-6

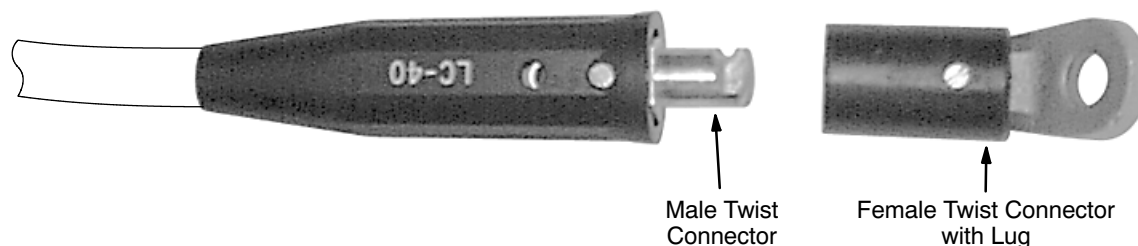
#### 4-4 PRE-RAMP PROTECTIVE GROUNDING CIRCUIT CONNECTIONS - PERSON 1 ( continued )

2. Disconnect the female twist connector lugs from the 3 ft. ( 1 M ) Power Supply Jumper Cable and then connect the lugs to the positive ( + ) and negative ( - ) power supply output terminals. See Illustration 4-7.



**FEMALE TWIST CONNECTORS ATTACHED TO POWER SUPPLY OUTPUT TERMINALS**  
ILLUSTRATION 4-7

3. Connect one twist connector lug ( shown in Illustration 4-8 ) of the Power Supply Jumper Cable to the power supply's positive ( + ) terminal. Make sure the twist connector is fully engaged. See Illustration 4-6, label "2".
4. Connect the other twist connector lug ( shown in Illustration 4-8 ) of the Jumper Cable to the power supply's negative ( - ) terminal. Make sure the twist connector is fully engaged. See Illustration 4-6, label "3".
5. Connect the .375 inch ( 9.5 mm ) lug of the 50 foot ( 15.24 M ) Power Supply Grounding Cable to the power supply's positive ( + ) terminal. See Illustration 4-6, label "4".



**TWIST CONNECTOR ON POWER SUPPLY JUMPER CABLE**  
ILLUSTRATION 4-8

#### 4-4 PRE-RAMP PROTECTIVE GROUNDING CIRCUIT CONNECTIONS - PERSON 1 ( continued )

6. Connect the Grounding Cable's .5 inch ( 12.7 mm ) lug to the magnet's grounding lug. See Illustration 4-6, label "5".

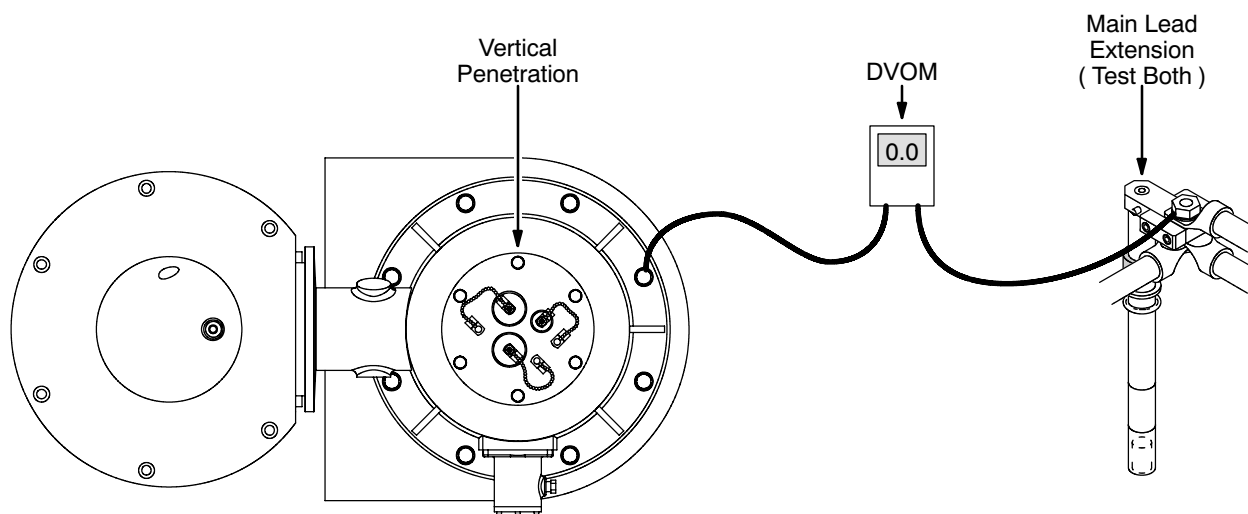
##### Note

If the magnet's grounding cable is connected to the RF Common Bus Bar at the Penetration Panel, the 50 foot ( 15.24 M ) Power Supply Grounding Cable can be connected between the power supply and the magnet or PDU ground wire clamp on the RF Common Bus Bar if more convenient. If the magnet is not grounded to the Penetration Panel, the Grounding Cable **must** be connected to the Cryostat Grounding Stud. See Illustration 4-6.

##### CAUTION

**Make sure there is a common ground ( resistance = 0 ohms ) between the Vacuum Vessel and the Main Lead Extension connections.**

7. Verify that the resistance between the Vacuum Vessel at the Vertical Penetration and each Main Lead Extension connection is zero ohms as read on a DVOM. See Illustration 4-9.



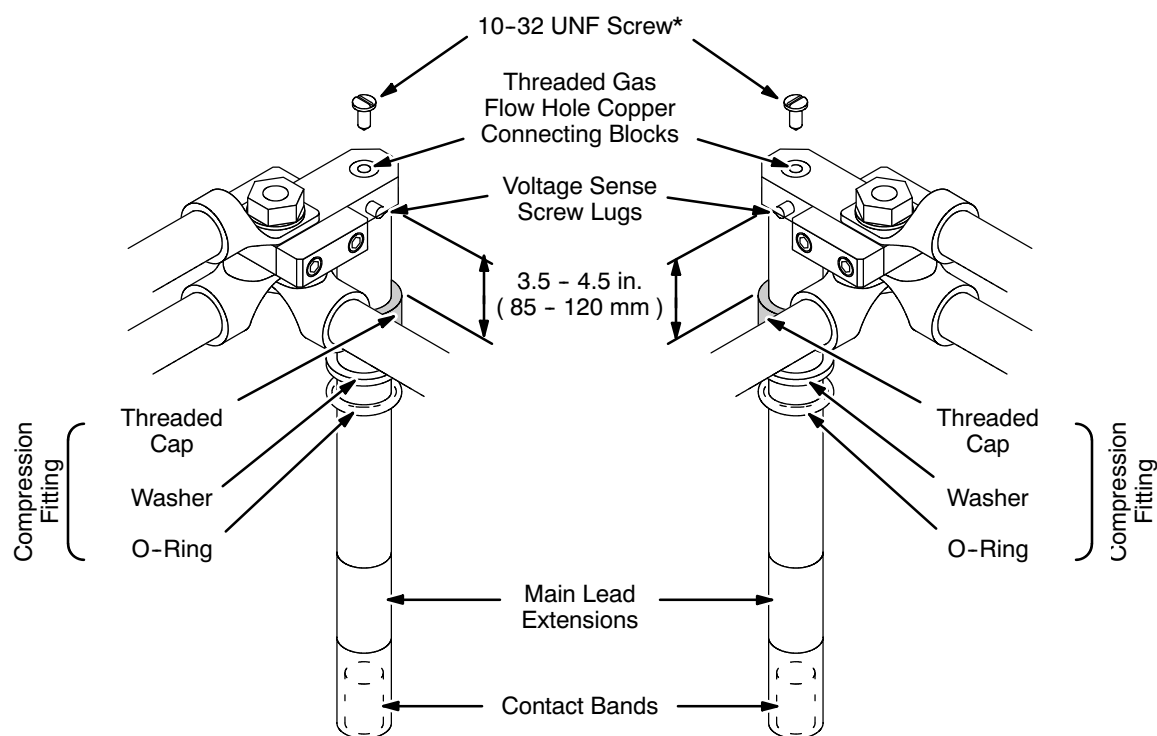
**VACUUM VESSEL TO MAIN LEAD EXTENSION RESISTANCE CHECK**  
ILLUSTRATION 4-9

#### 4-5 MAGNET CONNECTIONS - PERSON 1

1. Remove the threaded caps, washers and o-rings ( 46-294104P1, 46-294105P1, and 46-260389P1 ) from the plastic bag provided in the Magnet Ramping Equipment Kit ( 46-260703G4 ) and mount them on the Main Lead Extensions. See Illustration 4-10.

### WARNING!

**DO NOT LEAVE THE VENT SCREWS IN THE GAS FLOW HOLES FOR “DVM” CONNECTION SCREWS. THIS MAY CAUSE A QUENCH.**



\* Used to temporarily stop gas flow. Remove prior to ramping.

**COMPRESSION FITTING MOUNTING ON MAIN LEAD EXTENSIONS**  
ILLUSTRATION 4-10

**4-5 MAGNET CONNECTIONS - PERSON 1 ( continued )**

2. Place the end of the Main Power Cables on the Service Turret ( MSI-A3, A3 ). Allow about 3 feet ( 900 mm ) of slack.
3. Ty-wrap the cables at convenient locations on top of the magnet to avoid cable movement during ramping.

**WARNING!**

**MAKE SURE THE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON BEFORE STARTING THIS PROCEDURE. THE ODORLESS, INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE WILL DISPLACE OXYGEN AND CAUSE ASPHYXIATION UNLESS SAFELY VENTED AWAY FROM THE MAGNET ROOM. REVIEW AND FOLLOW SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.**

**WARNING!**

**THE CRYOSTAT EXHAUST SYSTEM HAS A 7.5 PSIG RELIEF VALVE, ALLOWING INTERNAL CRYOSTAT PRESSURE TO BUILD UP TO 7.5 PSIG. BEFORE REMOVING EITHER RAMP OR FILL CAPS OR LOOSENING ANY COMPONENT RESULTING IN THE RELEASE OF CRYOGEN PRESSURE:**

- a. **OBSERVE THE INTERNAL CRYOSTAT PRESSURE GAUGE READING,**
- b. **TURN VESSEL PRESSURE ALARM OFF IN 'CONFIGURE / ALARMS' TAB**
- c. **SET MAGNET MONITOR PRESSURE TO 0.5 PSIG, AND**
- d. **VENT THE CRYOSTAT DOWN TO 0.5 PSIG THROUGH VENT VALVE V2.**

**THIS WILL PREVENT EXPOSURE TO RAPID EXHAUSTING OF COLD HELIUM GAS DURING A SERVICE OPERATION.**

4. Check the Cryostat Pressure Gauge. If the pressure is above 0.5 psig, slowly open Helium Vent Valve V2 and allow pressure to drop to 0.5 psig. Then close valve V2.



**4-5 MAGNET CONNECTIONS - PERSON 1 ( continued )****WARNING!**

**A SUPERCONDUCTING MAGNET AT FIELD IS A HIGH ENERGY STORAGE DEVICE CAPABLE OF DISCHARGING RAPIDLY.**

**MAKE SURE ALL ITEMS COVERED BELOW ARE FULLY IMPLEMENTED WHEN INSERTING / EXTRACTING MAIN LEAD EXTENSIONS:**

- **MAKE ALL PRE-RAMP POWER SUPPLY PROTECTIVE CIRCUIT CONNECTIONS IN CONFORMANCE WITH SECTION 4-4, PRE-RAMP PROTECTIVE GROUNDING CIRCUIT CONNECTIONS - PERSON 1.**
  - **INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND ALLOW TO COOL BEFORE CONTACTING THE MAIN LEAD PINS.**
  - **DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOW THEM TO COME IN CONTACT WITH EACH OTHER.**
  - **WEAR NON-ABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
  - **DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
5. Remove the cap on one Main Lead Extension receptacle located on the magnet Vertical Penetration. Make sure the gasket inside the cap does not get lost.
  6. Immediately insert one Main Lead Extension into the open receptacle with 85 – 120 mm ( 3.5 – 4.5 inches ) of Main Lead Extension exposed. Loosely tighten the compression fitting.

**CAUTION**

**Make sure the Gas Flow Holes in the Main Lead Extensions are not blocked ( i.e., the 10-32 screws have been removed ) and that gaseous helium is exiting through the holes. Blocked holes prevent heat dissipation and result in high contact resistance.**

7. Repeat Steps 5 and 6 for the second Main Lead Extension.

**Note**

The Main Lead Extensions will depress approximately 1 inch ( 25 mm ) from the point of contact to the fully engaged position. A firm contact will be felt when fully engaged. Do not rotate the Main Lead Extensions excessively when in the engaged position as internal contact wear could result.

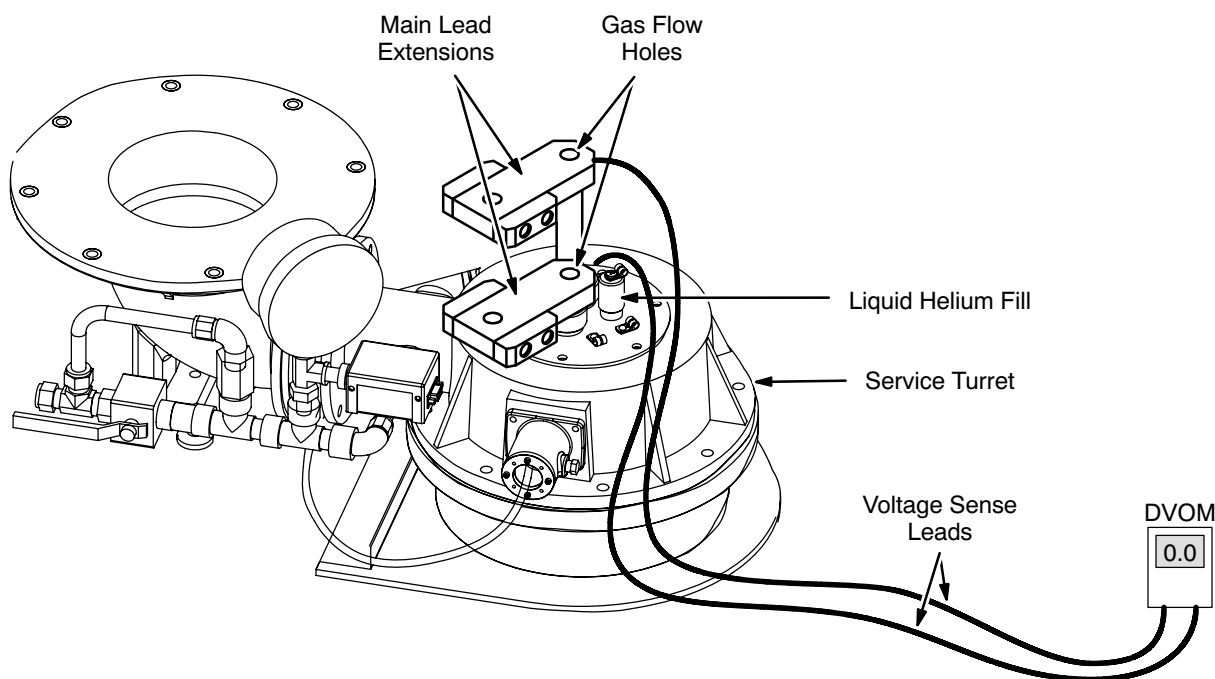
**4-5 MAGNET CONNECTIONS - PERSON 1 ( continued )**

8. When the Main Lead Extensions are sufficiently cooled, loosen the compression fittings and fully engage the Main Lead Extensions. Loosely tighten the compression fittings.

**WARNING!**

**CONNECTION POLARITIES MUST BE NOTED AND RECORDED IN DATA SHEETS, SECTION 5, MAGNET RAMPING & PARKING CURRENT LOG, TO PREVENT THE POSSIBILITY OF MISWIRING AND A RESULTANT QUENCH DURING FUTURE MAGNET RAMPING.**

9. Connect the Voltage Sense Leads to the Main Lead Extensions by clamping the spade lugs on the Voltage Sense Leads to the Main Lead Extensions with the screw lugs provided on the Lead Buss Bar. Terminate the other end of the Voltage Sense Leads to a DVM or DVOM placed near the Main Coil Power Supply. See Illustration 4-11.



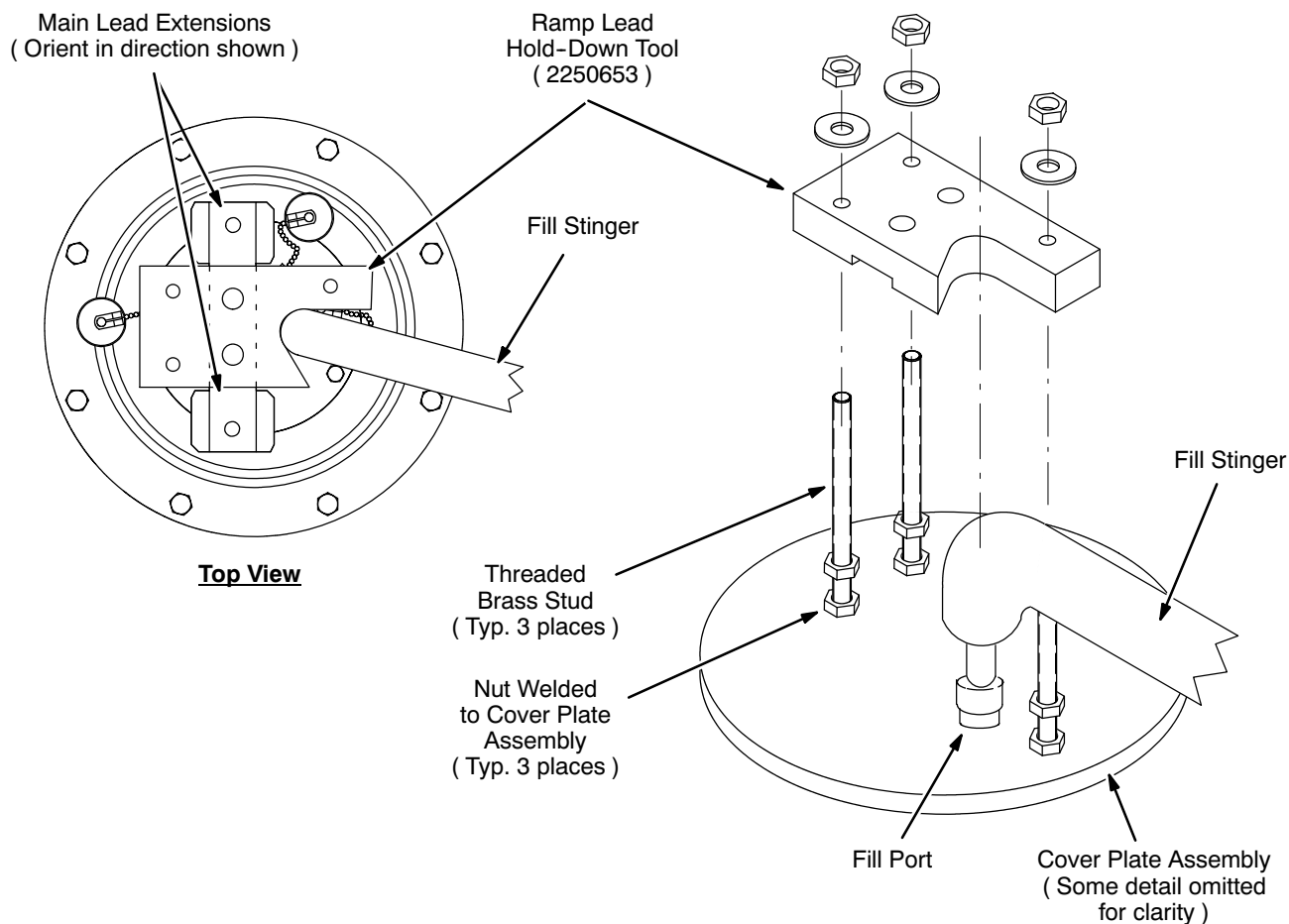
**MAIN COIL AND VOLT SENSE LEAD CONNECTIONS**  
ILLUSTRATION 4-11

**4-5 MAGNET CONNECTIONS - PERSON 1 ( continued )**

10. Insert three threaded brass studs into the nuts on the Vertical Penetration Cover Plate. See Illustration 4-12.
11. Install the Ramp Lead Hold-Down Tool ( 2250653 ). Make sure it is oriented properly, as shown in Illustration 4-12.

**Note**

The Ramp Lead Hold-Down Tool, contained in the magnet collector, will minimize Lead Extension motion and help obtain good contact resistance.



**RAMP LEAD HOLD-DOWN TOOL**  
ILLUSTRATION 4-12

#### 4-5 MAGNET CONNECTIONS - PERSON 1 ( continued )

12. Thread and tighten the top nuts to hold the tool in place.
13. Hand tighten the Ramp Port compression nuts.



**Make sure the Gas Flow Holes of the Main Lead Extensions are not blocked ( i.e., the 10-32 screws have been removed ) and that gaseous helium is exiting through the holes before ramping. Blocked holes prevent heat dissipation and result in high contact resistance.**

#### 4-6 TESLAMETER / PROBE SET-UP AND ADJUSTMENT - PERSON 1

##### 4-6-1 Set-Up

1. Mount and center the #4 Probe ( 0.35T – 1.05T ) in the magnet using a cardboard / wood box or a frame made from non-ferromagnetic material. See Illustration 4-13.

##### Note

Centering the Probe between enclosure surfaces will give sufficiently accurate magnet field strength readings for ramping purposes.

2. Connect the Preamp Box to the two Teslameter Probe inputs: the white Preamp Box wire to the Tesla-meter Probe left jack and the black Preamp Box wire to the right, lacing connector end of the Probe. See Illustration 4-13.
3. Turn on AC Power to the meter. The meter's rear view is shown in Illustrations 4-13 and 4-14.
4. Pull the "Lock / Manual" Switch and set in the "Manual" position.

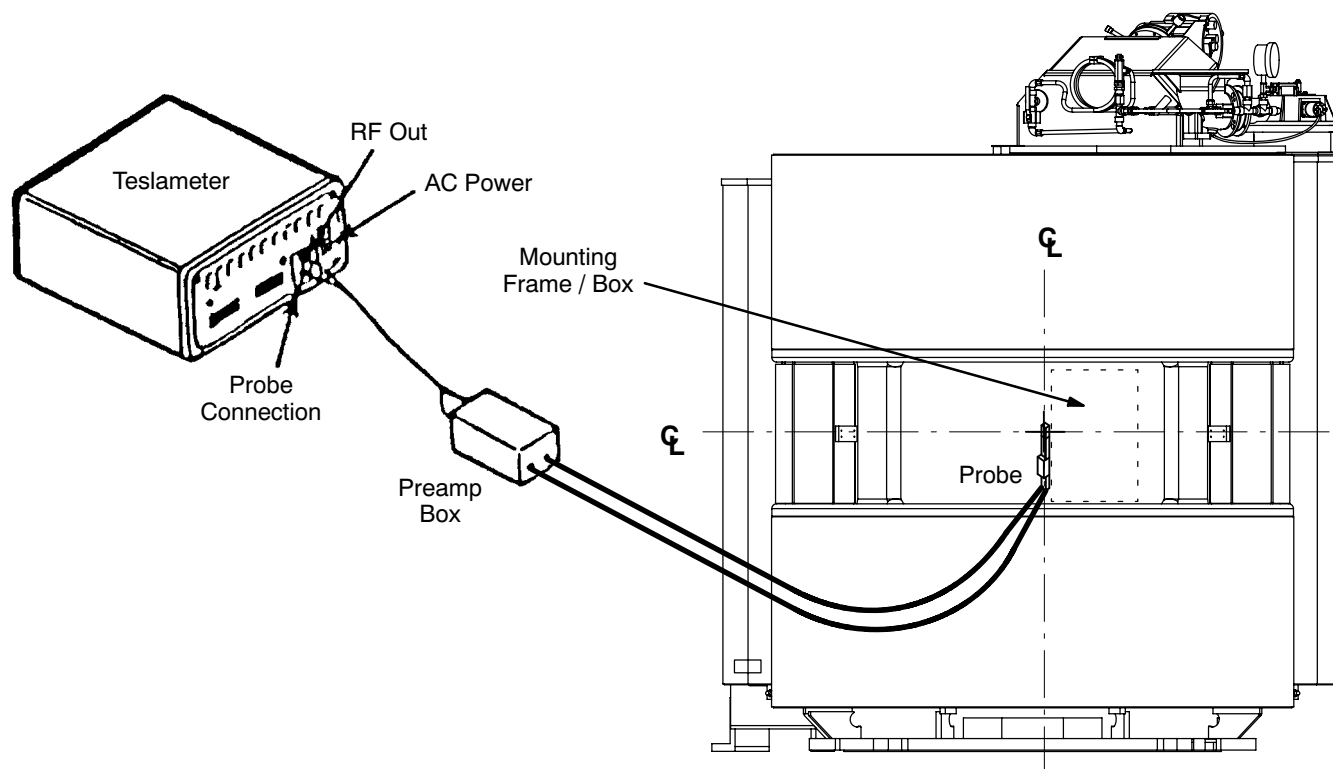


**If the Lock / Manual Switch is not in the "Manual" position at the start of field search during Ramping, the Teslameter's sweep will be in the  $\pm 5,000$  gauss range and the system could lock onto the mechanical oscillation harmonic produced by the Cryo-cooler Coldhead, resulting in an erroneous reading.**

5. Turn the Coarse and Fine Control Knobs fully counterclockwise ( CCW ).

## 4-6 TESLAMETER / PROBE SET-UP AND ADJUSTMENT - PERSON 1 ( continued )

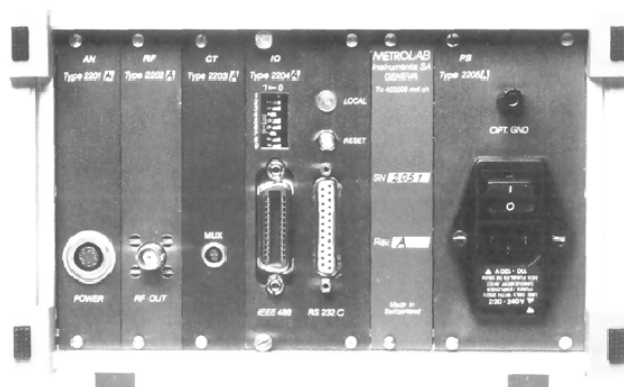
### 4-6-1 Set-Up ( continued )



PROBE SET-UP AND TESLAMETER INTERCONNECTIONS  
ILLUSTRATION 4-13



Front View



Rear View

TESLAMETER  
ILLUSTRATION 4-14

**4-6 TESLAMETER / PROBE SET-UP AND ADJUSTMENT - PERSON 1 ( continued )****4-6-1 Set-Up ( continued )**

6. Pull the “Freq / Field” Switch and set to the “Field” position. The Teslameter is now set up and ready to start monitoring the field when ramping commences.

**Note**

The Teslameter will not “lock in” on a probe signal until magnetic field is approximately 4000 Gauss.

**4-6-2 Teslameter Adjustment for Ramp Up**

1. Once ramping has started, monitor the power supply’s Current Meter until an indication of approximately 450 amps is approached. At this time, start monitoring the Teslameter.
2. As the developing field approaches the adjustable limit of the Teslameter probe ( approximately 4000 gauss ), the “NO STROBE SIGNAL” LED will start blinking. Watch for when this LED goes out.
3. When the “NO STROBE SIGNAL” LED is out, change the “Lock / Manual” Switch to the “Locked” position. See Illustration 4-14.

**Note**

If the signal does not lock in, change the position of the Probe Modulation switch.

4. As the field continues to increase, note when the “HI” LED to the right of the Coarse Control Knob begins to light. Turn the Coarse Control Knob clockwise ( towards the lighted LED ) until the LED is extinguished.

**Note**

As the magnetic field increases, the probe sample resonant frequency will increase above the range setting of the Teslameter. Therefore, increase the setting of the Coarse Control Knob periodically to keep the Teslameter locked on the probe sample.

**Note**

The Fine Control Knob does not function when the “Lock / Manual” Switch is in the “Locked” position.

**4-6 TESLAMETER / PROBE SET-UP AND ADJUSTMENT - PERSON 1 ( continued )****4-6-3 Teslameter Resynchronization**

If the Teslameter should go out of sync while ramping the magnet up ( or down ), it can be resynchronized by the following procedure.

**Manual Resynchronization**

1. Reposition the "Lock / Manual" Switch to "Manual."

**Note**

The "NO STROBE SIGNAL" LED will be on, and the "LO" and "HI" LEDs will be oscillating, indicating search mode.

2. Note the present current reading on the Main Power Supply's Current Meter.
3. Multiply the current read from the Main Power Supply's Current Meter by 6.85. ( There's approximately 6.85 gauss / amp ). Reset the meter at the resultant gauss level.
4. Slowly start increasing ( if ramping up ) the Coarse and / or Fine Control Knob while monitoring the "NO STROBE SIGNAL" LED.
5. Once the LED extinguishes, quickly place the "Lock / Manual" Switch in the "Locked" position. The meter will now be in sync.

**Note**

If the "HI" LED is lit, the Coarse Control Knob will have to be turned in the "HI" direction until the LED goes out. Repeat this adjustment as required until the parking field is reached.

**Manual Resynchronization Using an Oscilloscope**

An oscilloscope can be set up near the Teslameter to display and trigger on the "Field Modulation" signal from a jack on the Teslameter front panel. Adjust the time base to display one or two ramp wave forms. On the second channel display the NMR Signal from the front panel of the Teslameter.

1. Leave the Teslameter in the "Locked" position.
2. Slowly turn the Coarse Control Knob in the direction the field is going ( i.e., if ramping up, turn the control knob up to the higher numbers ).
3. As the meter is approaching the actual field, the baseline of the "FID" Display will start to wander. Once the meter is in range of the field, the "FID" will appear on the oscilloscope trace as the meter "locks on."
4. Once locked on, the "NO STROBE SIGNAL" LED will be out. Readjust the Coarse Control Knob slightly in the direction of the lighted "LO" or "HI" LED until that LED goes out.
5. Maintain tracking through end of the ramp sequence.

#### 4-7 RAMPING PREPARATION - PERSON 1

**WARNING!**

THE FOLLOWING REQUIRED SAFETY ACTIONS MUST BE TAKEN PRIOR TO RAMPING THE MAGNET:

- MAKE SURE THE MAGNET ROOM VENT EXHAUST FAN IS TURNED ON BEFORE STARTING THIS PROCEDURE. THIS IS REQUIRED TO EXHAUST THE ODORLESS AND INVISIBLE HELIUM GAS GENERATED DURING THIS PROCEDURE AND PREVENT OXYGEN DISPLACEMENT IN THE MAGNET ROOM. REVIEW AND FOLLOW SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.
- NOTIFY SITE ADMINISTRATION BEFORE RAMPING THE MAGNET THAT ALL MAGNETIC SAFETY PRECAUTIONS MUST BE TAKEN.
- POST WARNING SIGNS OUTSIDE THE 5-GAUSS ZONE TO ALERT PERSONNEL WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS AND OTHER BIOSTIMULATION DEVICES NOT TO PROCEED INTO THE DESIGNATED AREA. POST THESE SIGNS ON THE MAGNET ROOM LEVEL AS WELL AS AREAS BELOW THE MAGNET TO WHICH THE 5 GAUSS ZONE EXTENDS. SEE SAFETY REQUIREMENTS, 2301164PRE, MR MAGNET - SAFETY REQUIREMENTS.
- POST “RAMPED MAGNET” AND “AUTHORIZED PERSONNEL ONLY” WARNING SIGNS AT MAGNET ROOM ENTRANCE AND AT MR SUITE ENTRANCE, RESPECTIVELY, PRIOR TO RAMPING THE MAGNET. WARNINGS ARE TO ALERT PERSONNEL THAT NO FERROMAGNETIC MATERIAL OR INDIVIDUALS WITH CARDIAC PACEMAKERS, NEUROSTIMULATORS OR STEEL PLATES ARE ALLOWED IN THE MAGNET ROOM WHEN THE MAGNET IS RAMPED.
- REMOVE ALL LOOSE FERROMAGNETIC MATERIAL FROM THE MAGNET ROOM. PULL THE POWER SUPPLIES AS FAR AWAY FROM THE MAGNET AS THE CABLES AND SITE GEOMETRY ALLOW. METAL OBJECTS CAN BECOME DANGEROUS PROJECTILES IN A MAGNETIC FIELD. MAKE SURE THE MAGNET LIFTING SHACKLE, JACKING RAIL AND MOUNTING BRACKETS HAVE BEEN REMOVED FROM AND STORED OUTSIDE THE MAGNET ROOM.
- MAKE SURE THE MAGNET RUNDOWN UNIT IS INSTALLED AND OPERATIONAL TO ENABLE THE MAGNETIC FIELD TO BE QUICKLY DISCHARGED IN CASE OF AN EMERGENCY. SEE SET-UP AND CALIBRATION, SECTION 1-7.
- DO NOT WEAR STEEL-TOED SAFETY SHOES NEAR A RAMPED MAGNET. USE A FIBER / COMPOSITION TYPE SAFETY SHOE.



**4-7 RAMPING PREPARATION - PERSON 1 ( continued )**

1. Set up the field monitoring equipment, Probe and Teslameter in conformance with SET-UP CALIBRATION, Section 4-6, Teslameter / Probe Set-Up And Adjustment. Make sure the Probe is positioned at the center of the magnet.
2. Make sure the Magnet Power Supply is installed, checked and adjusted in conformance with FUNCTIONAL CHECKS, Section 8, ESS7.5-1000-2-D-1236 Main Power Supply Calibration / Check. Verify that the Heater 1 Main Power Supply is set for  $810 \pm 10$  mA.
3. Make sure the Input Power Cable for the Power Supply is disconnected.
4. Make sure the Ramp Power Supply is connected to the magnet in conformance with SET-UP AND CALIBRATION, Sections 4-3, 4-4 and 4-5. Also make sure the Ramp Lead Hold-Down Tool is firmly in place and cooling gas is exiting from Main Lead Extensions.
5. Record the Main Coil connection polarity in DATA SHEETS, Table 5-1, Magnet Ramping & Parking Log.
6. Set Heater 1 Main and Heater 2 Shim Axial Switches to the "OFF" position. Set the Current Adjust and Voltage Adjust Controls to zero ( full CCW ).
7. Connect the Input Power Cable to the Service Main Power Supply.
8. Make sure a Digital Voltmeter ( DVM ) is connected to the end of the Voltage Sense Leads and terminated at the power supply. The Voltmeter will measure voltage across the Lead Extensions of the magnet.
9. Make sure DATA SHEETS, Table 5-2, Magnet Ramp / Fill Profile, is filled out. Readings should be recorded at 3 minute intervals.
10. Disconnect and remove the Power Supply Jumper Cable from the power supply while wearing non-absorbant leather gloves with NO holes / tears. See Illustrations 4-7 and 4-8.

**Note**

Resistance measurements cannot be made and the magnet cannot be ramped with the Power Supply Jumper Cable connected.

**4-8 RESISTANCE CHECKS - PERSON 1****Note**

CCW = counterclockwise -----&gt; decrease

CW = clockwise -----&gt; increase

1. Make sure the Current Adjust and Voltage Adjust Controls on the Main Power Supply are off ( full CCW ). Turn on input power to the Power Supply.
2. Set Heater 2 Shim Axial Switch to the "I" ( on ) position to enable power supply output.
3. Set the Voltage Adjust Control to "400" ( 4 turns clockwise from zero ). This will set a maximum limit of 3 V ( ~ 0.75 V / turn ) for the power supply output.
4. Observe the Main Power Supply Ammeter and slowly turn the Current Adjust Control ( clockwise = increase ) to set 750A current through the Main Power Leads, Lead Extensions and persistent Main Switch.
5. Record the voltage reading on the Digital Voltmeter ( DVM ): \_\_\_\_\_ mV.
  - A voltage reading < 50 mV indicates the Power Supply Jumper Cable is still in place.
  - A 50 – 150 mV reading indicates acceptable internal contact resistance of the Lead Extensions.
  - If the voltage reading is > 150 mV, turn the Current Adjust Control off ( full CCW ), turn off the Power Supply Input Power and reseal the Lead Extensions on the top of the magnet.
6. Repeat Steps 1 through 5 if the Lead Extensions were reseated. Repeated readings > 150 mV indicate the need to replace the Lead Extension Contact Bands in conformance with REPLACEMENT / MAINTENANCE, Section 8, Main Lead Extension Contact Band Replacement. Upon passing the internal resistance check, continue with Step 7.
7. Set the Toggle Switch on the Power Supply Voltmeter to the "Main Power Supply" position.
8. Read the voltage across the Power Supply Output Lugs. A Power Supply output voltage of 2.1 V or less at 750A indicates acceptable system resistance. If the voltage exceeds 2.1V during the test:
  - a. Turn the Current Adjust Control off ( full CCW ).
  - b. Turn off the input power.
  - c. Check and retighten the bolts securing the Main Power Cables to the Power Supply and Lead Extensions.
  - d. Turn the Power Supply back on.
  - e. Repeat Step 8.
9. Upon passing Step 8, turn the Current Adjust and Voltage Adjust Controls to zero ( full CCW ) and continue with Section 4-9, Ramping – Person 1.

**4-9 RAMPING - PERSON 1****WARNING!**

- **MAKE SURE THE MAGNET IS 98% FULL OF LIQUID HELIUM AND A FULL 250-LITER OR LARGER DEWAR OF LIQUID HELIUM IS CONNECTED TO THE TRANSFER LINE BEFORE RAMPING TO PREVENT THE LIQUID HELIUM LEVEL FROM DROPPING DURING RAMPING TO A LEVEL WHERE A QUENCH MAY OCCUR.**
- **A RAMPING VOLTAGE > 2.5V ACROSS THE MAIN COIL OF THE MAGNET COULD CAUSE THE MAGNET TO QUENCH.**
- **IF A QUENCH OCCURS DURING RAMPING, IMMEDIATELY:**
  - a. **TURN THE MAIN COIL POWER SUPPLY VOLTAGE AND CURRENT CONTROLS TO ZERO ( FULLY CCW ),**
  - b. **TURN OFF THE POWER SUPPLY,**
  - c. **LEAVE THE ROOM AND**
  - d. **TURN OFF THE CRYOCOOLER COMPRESSOR OUTSIDE THE MAGNET ROOM.**

**DO NOT REENTER THE MAGNET ROOM UNTIL THE QUENCH IS OVER. A QUENCH IS A RAPID DISCHARGE OF THE MAGNETIC FIELD WHICH WILL RESULT IN THE RAPID GENERATION AND EXPULSION OF HELIUM GAS, RUPTURING THE BURST DISC IN THE VENT SYSTEM.**

**IMPORTANT !!!**

To prevent a quench if a Dewar should run empty, Person 2 shuts off helium flow and performs the following:

- a. Park the magnet at the present setting ( Steps 8 - 16 of this section ).
- b. Record the parking current and frequency in DATA SHEETS, Table 5-1, Magnet Ramping / Parking Log.
- c. Wait for Dewar change, Stinger plume and Stinger insertion into Fill Port V1.
- d. Turn power supply on ( heaters off ) and set parking current in power supply.
- e. Turn on the Main Axial Shim Heaters.
- f. Continue the ramp.

**4-9 RAMPING - PERSON 1 ( continued )**

1. Make sure:
  - A new ( full ) 250-liter Dewar is installed;
  - A 98% liquid helium level is being maintained;
  - The power supply's Current and Voltage Adjust Controls are set to zero ( full CCW ); and
  - The Axial Switch set to "I" ( on ).
2. Engage the Set Point Button ( depress and hold until the current is set ) and set the Current Adjust Controls for a reading of 800A.
3. Leave the Current Adjust Controls at setting in Step 2. Turn the power supply's Voltmeter Select Switch to the "Main Coil" position.
4. Set the Heater 1 Main Switch to "I" ( on ) and wait 3 minutes. Make sure gas flow is visible from Lead Extensions.
5. Ramp the magnet in conformance with Table 4-2 as follows:
  - a. Slowly turn the power supply's Voltage Adjust Control to obtain the Main Coil ( Ramp ) Voltage value shown in the table, starting with 2.50 volts.
  - b. Maintain the voltage over the current interval shown in the table.
  - c. Monitor the power supply's Current Meter. When the limit of the current range for the voltage setting is reached, slowly turn the Voltage Adjust Control to obtain the Main Coil Voltage for the next range.

TABLE 4-2  
RAMPING PROFILE

| "Stage" | Main Coil ( Ramp )<br>Voltage ( Volts ) | Main Coil ( Ramp )<br>Current Range<br>( Amps ) | "Stage" | Main Coil ( Ramp )<br>Voltage ( Volts ) | Main Coil ( Ramp )<br>Current Range<br>( Amps ) |
|---------|-----------------------------------------|-------------------------------------------------|---------|-----------------------------------------|-------------------------------------------------|
| 1       | 2.50                                    | 0 - 100                                         | 9       | 0.80                                    | 625 - 650                                       |
| 2       | 2.30                                    | 100 - 200                                       | 10      | 0.60                                    | 650 - 675                                       |
| 3       | 2.10                                    | 200 - 300                                       | 11      | 0.50                                    | 675 - 700                                       |
| 4       | 1.90                                    | 300 - 400                                       | 12      | 0.40                                    | 700 - 720                                       |
| 5       | 1.70                                    | 400 - 500                                       | 13      | 0.35                                    | 720 - 750                                       |
| 6       | 1.50                                    | 500 - 550                                       | 14      | 0.30                                    | 750 - Parking Amps                              |
| 7       | 1.30                                    | 550 - 600                                       | 15      | 0.00                                    | Parking Amps                                    |
| 8       | 1.00                                    | 600 - 625                                       |         |                                         |                                                 |

**Note**

Illustration 4-15, Magnet Ramping Profile, plots typical Main Coil ramping currents over time.

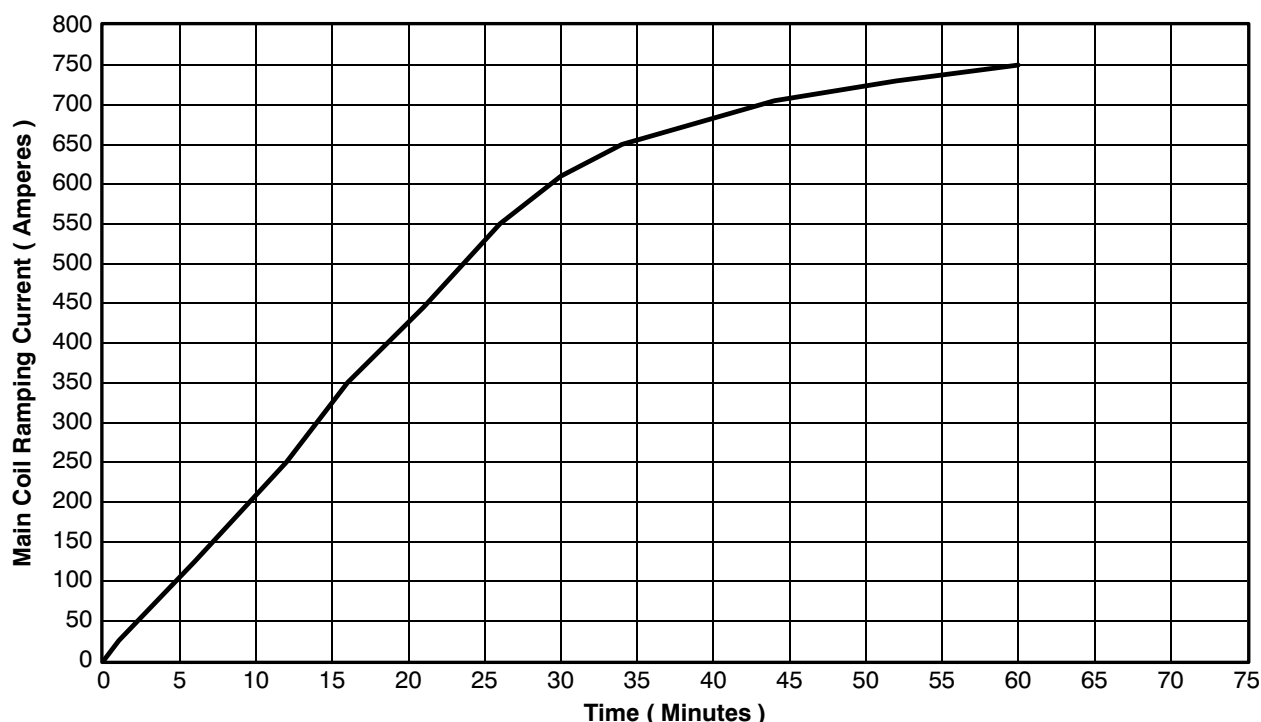
**4-9 RAMPING - PERSON 1 ( continued )****MAGNET RAMPING PROFILE**

ILLUSTRATION 4-15

6. When the Main Coil ( Ramp ) Current reaches 200 amps, estimate the system inductance by measuring current change over a 10-second ramping interval:

$$L_{\text{ ( inductance ) }} = 10 \times \text{Voltage} / \text{Current Change}$$

**Note**

This method will give inaccurate values of inductance when the current is less than 200 amps.

7. If the inductance ( L ) is outside the range of 5.3 – 5.5 henries, discontinue ramping, measure the main coil resistance in conformance with FUNCTIONAL CHECKS, Section 5, and contact your regional Magnet Team Engineer.
8. As the magnet field approaches the desired parking frequency ( 29.793 MHz ), adjust the Main Coil driving voltage to 0.00 V.

**Note**

The ramping field in step 9 was chosen at the low end of the allowable range because passive shimming will increase the field frequency. The system specification for frequency is 29.803 ± 0.01 MHz.

9. Check the Teslameter and adjust the Voltage Control as required to bring magnetic field to 29.793 MHz. This will account for the decrease in field after dialing the ramp supply down and the increase in field caused by passive shimming.

**4-9 RAMPING - PERSON 1 ( continued )**

**Observe the voltage displayed on the power supply Voltmeter ( toggle switch in “Main Coil” position ). When the switch goes “persistent” ( zero resistance ), the voltage across the magnet terminals will drop to approximately 0.00 V. This sudden voltage drop indicates that the switch has gone into the persistent mode. Make sure the switch is in the persistent mode before adjusting the voltage / current controls to prevent magnet rampdown to zero.**

10. Maintain the final current setting until the Main Switch goes into the “persistent” mode.
11. Set Heater 1 Main Switch to “O” ( off ). Wait a minimum of 7 minutes for the switch to fully cool and go “persistent”.



**MAKE SURE THE CONNECTION POLARITY AND FINAL RAMPING CURRENT ARE RECORDED IN DATA SHEETS, TABLE 5-1, MAGNET RAMPING & PARKING LOG. THIS INFORMATION IS ESSENTIAL FOR LATER CHANGING OF THE MAGNETIC FIELD. THE MAIN POWER SUPPLY MUST BE SET TO THE SAME CURRENT AND POLARITY IN THE MAIN COILS TO AVOID A QUENCH WHEN TURNING ON THE MAIN SWITCH HEATER.**

12. Record the final ramping ( parking ) current and lead extension voltage / polarity values at which the switch went “persistent” in DATA SHEETS, Table 5-1, Magnet Ramping & Parking Log.
13. After the switch goes “persistent”, slowly turn the power supply Voltage Adjust Control to zero over a one minute period ( full CCW ). Make sure voltage reading remains at zero.
14. Gradually turn the Current Adjust Control to zero ( over a one minute period ).
15. Set Heater 2 Shim Axial Switch to its “O” ( off ) position.
16. Turn off the Ramp Power Supply.
17. Disconnect the Input Power Cable from the Main Power Supply.
18. Reconnect the Power Supply Jumper Cable’s twist connectors to the power supply while wearing non-absorbant leather gloves with NO holes / tears. See Illustrations 4-6 and 4-8.

**4-9 RAMPING - PERSON 1 ( continued )**

19. Remove the Ramp Lead Hold-Down Tool and replace the bolts on Lead Assembly Mounting Plate.
- 20. Make sure the upper vessel helium level is at 90% and that the lower vessel helium level is at 100%.

**WARNING!**

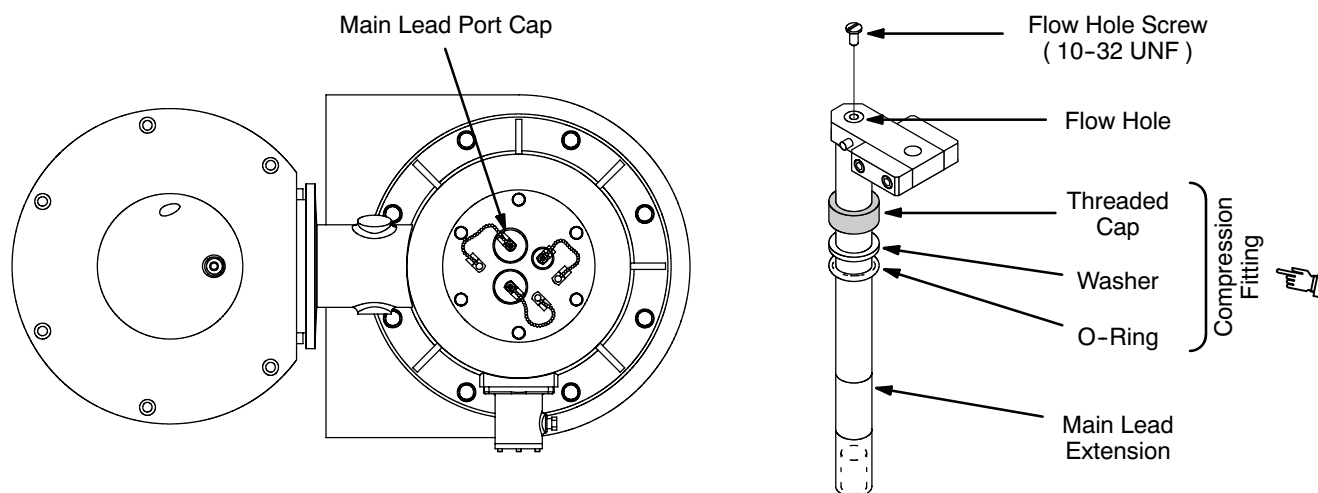
**A SUPERCONDUCTING MAGNET AT FIELD IS A HIGH ENERGY STORAGE DEVICE CAPABLE OF DISCHARGING RAPIDLY.**

**MAKE SURE ALL ITEMS COVERED BELOW ARE FULLY IMPLEMENTED WHEN INSERTING / EXTRACTING MAIN LEAD EXTENSIONS:**

- **MAKE ALL PRE-RAMP POWER SUPPLY PROTECTIVE CIRCUIT CONNECTIONS IN CONFORMANCE WITH SECTION 4-4, PRE-RAMP PROTECTIVE GROUNDING CIRCUIT CONNECTIONS - PERSON 1.**
  - **INSERT MAIN LEAD EXTENSIONS ONE AT A TIME AND ALLOW TO COOL BEFORE CONTACTING THE MAIN LEAD PINS.**
  - **DO NOT TOUCH BOTH MAIN LEAD EXTENSIONS AT THE SAME TIME OR ALLOW THEM TO COME IN CONTACT WITH EACH OTHER.**
  - **WEAR NON-ABSORBENT LEATHER GLOVES WITH NO HOLES OR TEARS DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
  - **DO NOT COME IN CONTACT WITH THE CRYOSTAT OR PLUMBING DURING INSERTION / EXTRACTION OF THE MAIN LEAD EXTENSIONS.**
21. After parking the magnet and disconnecting the Main Power Supply, plug and remove one Main Extension Lead at a time in the following sequence.
    - a. Open Valve V2 to de-pressurize the cryostat to 0.50 psig. Close V2.
    - b. Install the screw into the flow hole of only one Main Lead Extension at a time. See Illustration 4-16.
    - c. Remove all ice around the Main Lead Port compression fitting on the Main Lead Extension that is being removed ( i.e. the Main Lead Extension that has the flow hole plugged in Step 21b ).
    - d. Unscrew the Main Lead Port compression fitting and remove the Main Lead Extension from the magnet. Immediately replace the cap onto the Main Lead Port.
    - e. Repeat these steps ( a - d ) for the other Main Lead Extension.

**4-9 RAMPING - PERSON 1 ( continued )**

22. Remove the Power Supply Jumper and Grounding Cables.
23. Keep the Field Probe centered in magnet and proceed to SET-UP AND CALIBRATION, Section 5, Shimming Preparation / Field Stabilization.



**MAIN LEAD EXTENSION AND MAIN LEAD PORT COMPRESSION NUT**  
ILLUSTRATION 4-16